ActInf Livestream #023.1 ~ "Embodied skillful performance: where the action is"

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greetings hello everyone thank you for joining live or in replay today it is may 27th is that true not not even close not even close it's june 8th it's june 8th 2021 we're here in actin flab live stream 23.1 and we are discussing the paper embodied skillful performance where the action is so welcome everyone to active lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links that are here on this slide this is a recorded and an archived live stream with all of the advantages and constraints that that provides please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be trying to follow good etiquette for live streams from this short link you can access the calendar of events for live streams of multiple different kinds and highlighted there with the blue arrows are today's discussion on june 8th and next week's discussion that'll be on the 15th and they're both going to be on this paper embodied skillful performance so we had a good time making the dot zero and learning about this and kind of opening up what this was adjacent to and now we'll look forward to continuing the discussion and just seeing where it goes for those who are watching live it would be i guess especially appreciated to throw us some questions whether they're related to the paper or not the goal for today is to discuss and learn the paper is embodied skillful performance where the action is by hipolito baltieri fristen and ramstead from 2021 and today in 23.1 we'll be uh ambling skillfully we hope through the paper again and flagging areas where we want to be talking with the authors next week greetings dave welcome and just seeing where we get to opening up threads and kind of enjoying the middle part of the 1.012 sandwich so we can begin with the introductions and the warm-ups today um there's a few of us so we can go around in the introduction and just give a short introduction or a check-in and we can also then pass somebody who hasn't spoken taking a look at the warm-up questions so i'm daniel i'm a postdoctoral researcher in california and i'll pass to dean yeah hi i'm dean um i'm up here in calgary um and i'm getting more and more involved with this uh idea of sort of building up the active inference profile i'll pass it to david dave is there but if he isn't i'll pass it back to daniel dave you're welcome to join in whenever you would like i hear you tapping a little bit but maybe whenever you want to speak otherwise what is the active inference profile that's what i think we're trying to build out here i think we're trying to give that some anchors and some leverage so we haven't figured it out completely yet it's like everything in active inference it's it's a generative model it's growing and i don't know what its final form will be or if it takes on a final form even are there other areas that have profiles or are there other profiles of note yeah i i think there's convention that that's already an established profile and i don't know that this is has got to that place yet where it where we can say it has a convention or something that's predictable about it and i think that's what makes makes it kind of uh well in my mind it makes it curious where is it going as opposed to where it's been and what it what it's established so far nice points agreed active inference especially in these nascent times it's like a strange attractor where people come in from so many perspectives and then they get slingshot out in a different direction or with a different um memory or different view so dave no worries about the hardware mystery everyone is welcome to just type in the youtube chat if they have questions so i think for the rest of this stream we're going to walk through again maybe thinking again about walking as a skillful action and all the fun similarities and differences between motor actions which are what are discussed here and that sort of space where motor and knowledge come together like with skilled action yes dave your your uh different embodiments are muted indeed so the paper uh laid out their aims and claims and the central aim of the paper according to the authors is to discuss critically the limitations of instructionist control theoretic models of skillful performance so the big dichotomy or dilemma that's going to come up again and again is this difference between interactionism and instructionism so maybe dean where are you coming at that from or how would you describe the difference between instructionism and interactionism well well first of all even including interactionism as a formality i think is a huge step it's metaphorically speaking it's it's opening both of your eyes you have a second eye available but maybe you weren't aware of it and now with with the idea of turning that into sort of a a structured or formalized approach as opposed to just the instructionalism piece which is traditionally what we've formalized is it's not just a doubling of of what we might be aware of it's it's actually placing ourselves where we can see the depth of what we're observing or what we're analyzing or what we're addressing and i think that that's that would be a huge step if we could get people comfortable with that i'm not saying that people will necessarily get comfortable with it though because for a lot of people let's be honest um just following along and being instructed is difficult enough and now you're introducing a whole second lens of information and very quickly you'll have a bunch of people screaming no you're going to overwhelm them you're going well i'm not just because i have both my eyes open doesn't mean that i'm going to be i hope it doesn't mean i'm going to be scared i hope it means that i'm going to feel a little like a little bit more grounded and centered but we'll i don't know we'll we'll have a conversation about that today especially when we include the whole body yes the entire corpus one thought there instructionism being like instructions are passed from a to b you know water flowing downhill instructionism has a directional uh context and also the idea that what's being passed is the sort of packet of meaning like here's the envelope with your instructions now carry them out and what i think is exciting about interactionism is it's not just a bi-directional instructionism it's not like well you instruct me and then i'll instruct you that's sort of the the low bar for interactionism but where interactionism really blows the lid off is when we open into that third space of interaction and improvisation so i think that's what's kind of interesting to draw out and they are going to highlight in this paper how control theory models of skillful performance or otherwise implicitly or explicitly do have instructionist assumptions specifically this idea that motor representations convey or harness instructions about how to perform a specific task so even if the authors of those control theory papers didn't explicitly use the word instructionism then they're still playing into this idea that instructions are being passed for example from the brain to the body in order to be carried out and we'll be looking at that from a more formal perspective in a few minutes and the big claim that they're going to be reaching is that active inference doesn't need to posit that instructionist assumption so there's a way to frame control and skillful performance in the active inference framework that doesn't take on these instructionist assumptions and the content based directional uh message passing that it entails and that'll be something fun to explore also dave has a nice comment in chat dave wrote as i recall piaget and colleagues counted and described something like 400 distinct embodied concepts used by the walking skills of a typical 15 month old so that just shows how there's a lot going on even with uh behaviors like walking cool it's one other thing too the assumption that somehow uh the the muscles and the nerve endings sending signals that are in symbolic form is quite a stretch i mean we talked about that last week when we got to that inverse directionality slide but i thought about that a little bit after because i went back and watched after last week and i was just like i know for me i was incredulous a couple a couple of times we went through the paper with with blue but even going back and looking at it you'd have to you'd have to build a really powerful argument that says that the pressure signals on my fingertips are coming back to me in verbalized form exactly it's this idea that uh you know the files are on the computer like what's coming up from the fingertips is pain and what's going down is an instruction that's sort of the um cartesian dualism mapped on to is just like body and brain are two disjoint pieces and they pass special symbolic information to each other right and we'll see our brain and our bodies having a conversation so what is it if not a conversation or could it be beyond a conversation right right nice cool we covered the abstract last time all we'll point out here is just that they start as good philosophers with an examination of the instructionism and what that entails the second and third sections of the paper are characterizing the representationism representationalism that's related to these motor commands kind of what we were just alluding to and a special focus on the optimal motor control theory and then the final sections take it to predictive coding and to active inference which are formal frameworks for modeling behavior that are related to control theory formalizations but the goal of the paper is to show that some of the instructionism is left at home which allows us to go out with interactionism here is the roadmap framing a similar uh outline and also just giving a few more uh signposts along the way with the figures and the box and whatnot the keywords provided here were helping us understand what did the authors want to have their work indexable near in the semantic net of literature online how did the authors want their work to be grounded and it was in skillful that's in the title optimal motor control that's the sort of um scapegoat or foil that's the contrast maybe even bringing us back to contrasting divergences instructionism again tied up with the optimal motor control motor representations lending us into action-oriented representations and finally taking us to active inference where we begin and end so skillful performance can mean many pieces and the authors defined it as opposed to bear movements such as breathing and blinking skillful performances are intelligent bodily activities which harness knowledge about how to perform certain movements expertly it reminds me of the eternal debate who slash what is intelligent if an insect knows how to walk or if it is trained to do something versus if it's something that its body provides the training for inherently what is skillful so definitely will want to ask the authors and anybody in the chat and anybody who joins us next week what is skillful performance what is unskillful performance if somebody tries to do a skillful performance and fails is that still within this category of behavior i saved i saved up for this week because i wanted to get to the after the comma the witch harness which harness knowledge uh like i kind of hinted that that's that i wanted to ask a question and i still do want to ask a question well what does that mean to harness knowledge um consciously or or is the can you get to a place i mean blue talked about this you talked about this can we get to a place of doing it automatically does that mean it's harnessed just because we're not thinking about it and and how much time did they spend looking at that piece because that that kind of demarcates out what skill is is it if it's a dependent if it's a dependency on harnessing then what how do we define that how do we define what's what's captured and what isn't and the harnessing brings up kind of like harnessing or yoking uh you know buffalo or a horse or something like that and that's so kind of hinging on that word it's like the knowledge is that wild stallion and it's actually the bodily activities which harness that knowledge because when you quote know that your body can move in many different ways but skillful performance is when that is harnessed it's when it's connected to an instrument or whether it's its own instrument and it's channeled in a way maybe that's productive or expressive well at a fundamental level we have to again be careful that that something going on at our fingertips is being harnessed by something going on three and a half feet away yes connected but but is that an assumption that's verifiable is there an actual harnessing going on even because lots and lots of times things happen at my fingertips that i'm not sending i'm not even aware of what my hand is doing my wife tells me stop jiggling right just because i got nervous energy or whatever i think that's going to come back also to the predictive processing and predictive coding frameworks which is sort of when the body is getting the same the signals it expects like if you're not consciously bringing your attention to you know your foot if it's just your foot in the shoe it feels like nothing or there's no attention paid to it and you can kind of shine that light of attention on it and it will feel normal or maybe you can start to zoom in on other sensations like tingling or other feelings but the sort of default state when you're not paying attention things that are happening the way that you expect is neutral and so that's what that's why that active inference piece is really interesting because in order for it to be active inferencing there has to be some kind of attention or awareness being paid and that's like it's attention as um a resource gotta pay the cost of attention or suffer suffer the regret of not paying attention and dave dave wrote trying and failing especially if part of the problem is choking which was brought up by the authors in the paper as well is likelier to involve more neocortical activation and more access consciousness than smooth performance of the same routine so trying and failing there's a lot more inner dialogue or thought process like awareness happening whereas when something is performed skillfully and smoothly it just sort of goes off without hitch and just you know you whip out the instrument make your your shot or your swing or whatever it happens to be and then maybe you can narrate in retrospect but in the moment you're not having as much awareness so that's something one last voice oh yes so this is really interesting to me because again um is it skillful to be in the peloton and be be pulled along by what we would describe as the intention of those in front of you is that considered to be a part of this because there's a lot of things that we're aware and conscious of and some things that happen in these skilled performances they're just because they're part of the collective they're they're caught up in the inertia of the moment and is how do we how do we factor that in when we're talking about active inference because i think that's what instructionalism [Music] doesn't attend to there's many many things that are going to get dragged along here just because they're drafting and how do we how do we address that when we're talking about skill performance skillful performance and interactionism because interactionism at least says i can be caught up in an in an inertia moment i don't have to explain it absolutely um instructionism has to pitch and hold everything into that symbolic information theory what is the instruction from the front of the pack of the flock of birds or the peloton of cyclists what is the instruction that's getting passed back and so we have to go from what we know is just happening which is like the physical inertia there's also a narrative inertia there might even be just an air current going over the person that actually stabilizes them in their um spot we don't need to fall back to referring to those as instructions when they're clearly interacting phenomena different types of interactions you could have uh somebody breathing heavily next to you and that could give you a signal but it's about the unpacking of that interaction and then where that takes the collective rather than again just thinking about everything as connected with wires and passing secrets right cool um we can look again at the active infrared plant so this was just asking where are the limits of skillful if the niche and the fitness to the niche is what counts for different organisms our human cultural niche has all these awesome kinds of skillful performance and otherwise but what about other creatures what about other phenomena the kinds of things that we might want to also model with active inference but just don't quite have that same level of um maybe thinking through other minds as we have with people whereas we try to put ourselves in the shoes of somebody else do you think daniel that that plant has um got an embodied technical skill welcome blue hey blue hi we're just wondering whether plants have the um this plant okay but i have a technical skill now that it has the the tools to be able to move from side to side it's a great question i wonder if if it could be shown that it got better at moving from side to side that would start to look increasingly skill like and i think it's it's that idea that it could uh the molecular changes could occur that would allow the plant to respond faster or you could imagine selecting on a whole greenhouse of plants and the ones that are able to switch faster those are going to grow faster and then you select the ones that grow faster based upon their ability to learn in that generation so it kind of shows how you get this development of skill within a life that's development but then between generations you can also have the um entrenching or the canalizing of learned skill like language and skillful performance in in this paper and in our slides it's kind of like we're hinting a lot at this full body type performance uh high jump and things like that but of course language is maybe one of the examples of skilled performance it's a motor behavior it involves a huge number of interacting motor subunits voice box and the mouth and speech therapists and others they go deep into that but speech is almost like the case where you say well surely instructions are being passed that's maybe the the high water mark for where interactionism could go would be to have a speech a framing of speech that moved us beyond instructionism because this seems like clearly the transmutation of semantic or symbolic information into not directly related motor behavior so that would be kind of a nice case to investigate and dave wrote in the chat it seems odd that dogmas that have been known to be false for decades are still premises of so much funded professional activity i'm thinking of bf skinner's insistence that all control activity must be strictly cortical even though it's been known since at least the 50s purely on the basis of speed of nerve conduction that that would prevent graceful walking let alone faster and more intricate movements so that's pretty interesting we have almost just anatomical reasons to know that not everything can go through this sort of control and cognition in the loop there's just not fast enough transduction along the nerves so there has to be this sort of distributed function more consistent with an interactionist framework than an instructionist framework yet it's easy to fall back to instructionism maybe especially for professionalized areas let's go to um optimal control blue if you wanna you know feel free to add anything that you'd like optimal control theory is a nice bridge point because in the paper and as we'll be going into we're again thinking about motor behavior but control theory is something that applies to the electrical grid to logistics to robotics which are i guess motorized but optimal control theory is a framework that expands to many non-motor systems as well so that makes me wonder about whether we can also apply that instructionism versus interactionism contrast to control theoretic situations that are implicitly instructionist like there's an algorithm and then it sends the buy order to the unit that makes the purchase on a market or it sends a certain piece of information to a different computer that then enacts that like a suggestion and then even in systems where we know instructions are being sent like computer systems maybe there's an interactionist way to think about that too any other thoughts or does optimal control essentially place all of our chips on the top down uh aspect of of the relationship between top down and bottom up does it does it do it does it optimal say optimization happens exclusively from a top-down directionality and that there is really no place for the bottom-up part i think it's a good question and i'll skip to this fristen 2011 paper which is entitled what is optimal about motor control right i wonder if um that optimality does arise from the top or um what is optimal once we move beyond instructionism there's a perfect set of instructions there's a perfect way to do it and then that is either carried out faithfully or it's not that so philosophically if you said optimal would be some some um openness to both an ability to recapitulate and an ability to discover to be able to put those two things into some sort of coherent relationship um i'm not saying that that's what is being said here but well i'm just not sure i'm not sure why we always assume that optimal is always about being able to create copies even in scope of skillful skillful even compared to a standard there are things that happen when a gymnast does their routine where some some element of the the variability that applied to that situation is considered when something is being judged i think the only thing that that doesn't doesn't factor in variability is the clock right like you either ran 9.8 seconds or you didn't but many of the skill skills skillful performances that we pay attention to aren't always about a recapitulation so it's interesting i don't know i i guess again it's how you oh how you define it but we want to we actually want to get on to doing something useful here so yep and also here's a line in that first in 2011 that reminds me of dave's comment which is uh these estimates which are coming from a predictive processing or active inference framework can finesse problems incurred by sensory delays in the exchange of signals between the central and peripheral nervous system so the generative model can include space for that delay when we think about just sort of two people who are talking and accounting for each other's delay for example in that kind of an interaction versus the send process weight send back model there's actions that happen faster another example that maybe would be like baseball where there's such a small amount of time to decide whether to initiate the behavior of swinging and this paper from 10 years ago laid out just such an difference between value learning and active inference which allows for pragmatic and the epistemic component the optimal suggests multiple things first off it's like it kind of suggests that there is an optimum that is being achieved so that's one maybe easier way to one easier point just say okay well maybe there's an optimal way maybe there's not an optimal way but what is being optimized is the value just like reinforcement learning what's being implicitly optimized here is a value function and just as it says here the replacement of value and cost functions with prior beliefs about movements removes the optimal control problem completely so this idea of the value and cost it's kind of like an economic analysis of motor behavior so then let's just imagine that any number of circumstances like somebody um has a belief about how a performance will be viewed socially well you could couch that in reward and you could try to go break down from the narrative and the social all the way down to the movement of the fingers on the piano about the value and well if you mess up it's a lower value action but you're getting really far away from the embodiment of the performance which maybe doesn't have that same sort of cost all the way down so it's like will we have cost functions and economics all the way down and value and cost or will we have generative models and an instrumental way of thinking about action blue anything overall otherwise also anyone who's watching lives welcome to ask a related or an unrelated question i'm good anything after last week's discussion that you'd been thinking about um i'm trying to remember last week's discussion refresh my memory yep that's what this first part is let's go to figure one and two of the paper and take a more formal look at uh and we can go to first in 2011 if we want more details on the similarities and differences because this is kind of the core of the contrast between opt optimal motor control theory and active inference it's going to come down to the differences in how these models are framed because there's an interpretive layer but at the heart of it it's the differences between these two kinds of models between figure one and figure two here so in figure one uh and we'll go to the juxtaposition in figure one and figure two we have in both cases the plant kinetics which are defined with the exact same equations and a sensory mapping so this red box is the same between both of the models between control optimal motor control on the left and active on the right and there is still an optimal control module in active inference so that is interesting but if we look a little closer we can see that in figure one that uh motor commands that u with a tilde over it it's the minimization so it's optimization minimization maximization sometimes by minimizing the cost you maximize the profit or it's flippable throw a negative sign in there and a min becomes a max so here what's being minimized is an integral of the potentially the cost function c of x and u for changes in time so cost is being minimized through time and that is optimal control in that framework when we look at the optimal control in the active inference model we see a really different structure even of that optimal control module we see that u which is playing a similar role it's not a command but it's still being passed this u of t function is still related to a minimization but what's happening inside of the equation is very different and we can um go to the first in figure caption to see what is happening there that e curly e is the distinction between exteroceptive which is e with little e and proprioceptive e with a p reporting on hidden states in intrinsic extrinsic and intrinsic frames of references respectively these prediction errors are the difference between sensory input observed and predicted so going back to here what's being minimized isn't the cost function through time even philosophical disagreements with homo economicus aside okay let's put aside cost and then then there's the challenge of assigning costs to different kinds of states what's being minimized here is the cost through time i mean don't businesses try to do that everyone wants to figure out how to minimize cost or time it's challenging to do what's being done here the minimization between the proprioceptive sensory differential with a multiplication potentially through time not sure what that does in the model but it's just so interesting that um the minimization here is just like we've talked about in so many of these discussions it's about your generative model of action and sensory outcomes sensory outcomes are emitted by hidden underlying states that you can't directly observe like you don't know exactly whether it's night or day but you are getting photons or not so there's a hidden state that's being evaluated that's the sort of hidden markov model component and the hidden state is in the model emitting states that are being observed as sensory outcomes and in active inference the depth of that generative model is actually such that instead of having a really rich generative model of cost functions and then trying to reverse engineer your cost function to find the cheapest behavior the generative model is of expected sensory outcomes like every time i move my shoulder here it starts to hurt and we don't need to introduce a concept of reward or cost into that equation we can suffice to say that there is a generative model of the sensory outcomes under different motor policies and the organism has a preference for let's just say being in non-painful states it sidesteps that question of cost in a way where we can see that even from the side by side even where there is an optimal control element like a minimization element what's being minimized is something that is very very different and that's one of the key aspects of active inference that what's being minimized is the divergence between expected outcomes and realized outcomes that's what of course makes it so related to predictive coding predictive processing rather than drawing only on the reinforcement learning type approaches that are doing a implicitly maximization on reward daniel can i color in a little bit too please when i saw this it was interesting because what i saw was a rep not i don't i guess it was a replacement so on the figure one it's do estimation and then in figure two what i saw was the removal of that and the application of a rule so uh minimize uh prediction errors that's the rule as opposed to a step and all i thought about was for example hippocratic oath i i don't know what i should be doing in terms of this patient but the rule is do no harm so i think that's what i thought was really interesting here is that an actual step was replaced with a rule and that and that to me was the big differentiator the introduction of rules that's why i asked last week when you changed the rules of the game for the plant i know the affordance has changed and i know the tools were introduced and all of that stuff but i i keep coming back to tools rules and pools tools being obvious rules being obvious pools just being an aggregation of resources and so i think if you if you're able to sort of coordinate all three it gives you a different sense of what might be going on okay a fun thought on that just to think about game playing the rules are always in effect and the rules can be explicit or implicit like it's a rule in chess that you have to move out of check or figure out how to get out of check but then there's sort of what are equivalent to rules like if a piece is pinned it can't move but you don't need to state that as a rule so rules are they're simultaneous they're always in effect unless there's another rule and they can breed emergence you just say okay there's here's how the pieces move and then here's check those are the rules and then it's as if there's a rule that is you can't move out of a pinned piece so rules always in effect allow for the space of emergence steps have to be sequential and so it's almost like we're moving from the um within a round of a game that's the instructionist okay you're playing risk first roll the die now see whether the three is higher than the two die or whatever it happens to be and interactionism is pulling us back to like the rules in the pattern that still there can be sequential instructions or sequential information can occur but it is very different to have rules rather than specified steps so let's see we're building up yeah go ahead i just want one last thing so when i would ask young people to watch a video clip of something and i would ask them to draw three columns and title them tools rules and pools and find examples find tokens of each what do you think was the easiest thing for them to accumulate under it was tools yes those those have got the edges what was that what was the second highest aggregation pools because they could they could literally figure out what was congregating what was the hardest thing the parallel processing which is what you just described it's always going off the sequential is much easier to account for the parallel processing the happening as co-occurrence is the hardest thing for people when when you're walking into a novel to take account of so that's you can you can say it it's just being aware of it and finding examples of it that's really really hard for most minds so even from a computational aspect right like the parallelizing functions make them like speed up so much right but it's they also like are so computationally intensive right yeah exactly dave has another great comment on this slide he wrote um i'm looking for where time is applied in the two approaches the intrinsic frame of reference for active is u tilde of t so that's um we'll zoom in a little bit on just figure two here's that u-tilde of t passing through the intrinsic frame of reference um but there's no explicit t in the optimal control formula we can just see that motor commands utilda are passed so there's kind of like a timelessness to con you know contract this muscle it's not contract this muscle now it's actually just um a timeless statement so t he says whereas t is inside the optimal control formula um here is t uh dt it's a derivative through time and not in the intrinsic frame of reference as seen in active so he asks whether that tells us where the approaches are taking account of time differently i think that's a great question is how does time play um a role differently in these two frameworks let's keep on looking for similarities and differences so we have rules not steps for active we introduce a time dependence of that downward motor signal um we have of course a deep generative model component we have state estimation is that's the um the hamlet joke there's this whole module that's clearly different in optimal which is the state estimation and that's what doesn't need to be done in active inference is this state estimation of okay given the sensory input we're getting like i'm getting um you know two-thirds of my proprio receptors here and one-third here are firing so let's estimate that the arm is here okay now pass that to the next part of the model given that the arm is here what do we want to get the arm to okay now given which side of where we want the arm to go to we estimate the optimal and the actual r send the command you know turn the steering wheel to the side to bring it into alignment with the optimal or the most rewarding state so state estimation is in optimal motor control theory but an active inference you just don't have it and that's sort of an interesting piece to drop out of the model because we don't need to have an explicit representation or an explicit variable with the location of the arm so that's a rule it's like john cage everybody has the best seat it's almost like it's enough to just say you're on the ocean you're sailing and then you're going to be trying to reduce your uncertainty as you pursue policies that in your generative model of the world which includes latent causes so the degenerative model includes like latent causes of the world which are not anywhere to be seen in um optimal motor control theory like there's a deep latent cause that if too much weight is on you know this branch it's going to break where is that the constraints of the niche or something that's deep detonation not there so anyways it's like you're in the boat you want to be reducing your uncertainty about getting the sensory states that you want like seeing that land look bigger reflecting you being closer but you don't need to have that gps coordinate where am i okay now where do i want to be and how are we going to get there it's enough to actually have a sensory preference and then reduce your uncertainty on the trajectory of reaching that sensory outcome so that's a big difference in active inference and i'm seeing even a few more let's look at the green boxes so in the green box for figure one we have a forward model so they write that the function of the forward model is to improve the execution of action it's kind of funny executive executive office or the executive component of the government it's about executing that's where the instructions are issued from executive orders improve the execution of action by helping to finesse the inferences of the state estimator so again it's in feedback with the state estimate and then it's sending an effort copy so that's outgoing is efference and so there's this feedback between the prediction of um basically where the body is in this case and then that gets sent it's kind of interesting that the x comes from the state estimation that's the where is the arm and then the forward model is the forward model of motor control that is what connects with this u tilde that's the green box that forward model in optimal motor control whereas in active inference we have a different forward model it's a forward model of x as well as v which is the prior belief so again there doesn't need to be um a forward model of position and motor action as happening here it's enough potentially to have a model of action um and over beliefs we can look at the first in um caption to see a little bit more closely uh yeah any thoughts on that i think the state estimator to go back i think one live stream state estimator says that there have to be ties between the two rails to stabilize those ties so that some processing can take place i think what this talks about is the fact that we know that there's a gradient we know the train is going to roll downhill as long as there's a gradient it's a rule as opposed to trying to stabilize the relationship between the ties of the railroad two very different ways of looking at something and i think um this goes kind of back to what i said i think there are times when we do give ourselves instructions and then there are times when we simply parallel process because we have a rule i don't want to make mistakes because that doesn't serve my my goals or my target and i think all i i again i just suggest that the inclusion of both is probably to our advantage it's just that what we tend to do is give all of our we've been trained to give all of our attention over to making sure that the thing is stabilized nice also one piece of first in 2011 that's not brought over to this paper figure one was that motor control figure two is predictive coding and so already we see the reduction of the model in the terms of the state estimate goes away you don't need to explicitly have a state estimator because the state estimator has now been subsumed into the forward model which is why crucially the mapping from hidden states to sensations is now part of the forward generative model so we have a generative model of what we expect and now with predictive coding instead of keeping a variable of what we think is happening we just have which is can be tracked through time and all that we're simplifying that state estimator by including the expectations about the state estimator in our forward model that's predictive coding how does that differentiate from active inference so that's a good question here we have plant kinetics sensory mappings forward model and optimal control with a cost function still coming in you still see that integral with the of motor commands u over time whereas in active inference again we're moving from it's more valuable to be getting the signals i like and i like rewarding things to i'm just bringing the sensory outcomes into alignment with expectations and i'm selecting policies based upon which ones i think are going to get me there and then this section that um i had highlighted in summary the tenet of optimal control lies in the reduction of optimal motion to flow on a value function like the downhill flow of water so that's pragmatic value and that's awesome because think about how much work has been done on information flows in the last 10 years conversely in active inference the flow is specified directly in terms of the equations of motion that constitute prior beliefs like patterns of wind flow the essential difference so how do we go water going downhill on the value you know gradient descent or up and down again they kind of get flipped around but how do we go from just thinking about what direction does gravity take you that's value that's reward there's one answer you know which policy for financial outcomes is going to have a higher value that's that scalar estimate of value the essential difference is that prior beliefs in active inference can include solenoidal flow so this is where we're opening up the whole discussion of play so i hope that we can think about that for a second and fristen says you know he doesn't want to overstate the shortcomings of so it's not that you can't introduce a solenoidal component to value you can have the idea of solenoidal flow circulating on iso contours but how many businesses stop and say you know let's not pick the most rewarding number let's look for maybe it's not the most rewarding number but then there's the most solenoidal play around that number people say but we're not going to be doing those so why would we stop at you know 5 000 feet when we know that we can get to 6 000 feet and solenoidal flow and the ability to have pragmatic and epidemic that directed and the solenoidal components opens up the space of play hopefully so what do you think about that or does i'll put up this slide where we had separated out this optima motor control theory with a value only that's the one thing that this robot is tracking is the slope and the altitude it's following is uphill as much as it can so it can get to the top of mount value whereas in active inference we're using the helmholtz decomposition which is related to the physics of flow on vector fields that is going to reduce to these two irreducibly separated components so like they're totally orthogonal they don't have redundant information they have totally separate information there is the value component the straight line but there's also this equal value any thoughts on that you explained it really well here's a comment from dave if while people are thinking um giving oneself instructions can refer to something real and useful if one interprets the execution of the instruction appropriately i'm referring to playing a piece of music without touching the instruments which can be nearly as effective as actually playing if the challenge is to simply touch the right stops at the right moments this can actually be more affecting empowered machine operators may seem less efficient than highly regimented ones but may over time get to far more effective overall operations dean what do you think about that again i completely agree okay yeah so my my only question is so why more people not picking up on the potential of this and the value of having both an awareness of both so i think like it's maybe just getting started in a computational way i mean we've we've explored some computation papers um looking at you know how to optimize a given task computationally so i think that it will um gain traction right it's just so new i mean not active inference as a concept but um the practical applications of active inference i think are um just starting to be explored and i think that the play um aspect is is important i mean because it's it's like how we learn right exploring or playing or or um you know these types of things give rise to to real learning uh even if it's accidental learning like necessarily effort directed yeah i think i think it's also i think it's something some has to do with um a perception that play doesn't necessarily generate value which is kind of sad because there's so many examples where it does the outcomes of play caused your mind to go off in so many because it's supposed to be about distribution this is a kind of a way of of of encouraging that distribution and yet for some reason it just doesn't get the same respect so i'm just thinking about the um pipeline from basic to translational to applied to sort of infused research like once oh everybody knows that you know that what what was once basic though becomes applied and then becomes every day so let's just imagine that we're going to shooting on that skilled performance slide well if we have implicitly or explicitly if we're working under this optimal motor control function we're thinking okay maybe people's biomechanics differ but for each person's biomechanics there is an optimal way to shoot a free throw and we want people to have a more accurate modeling of where their hand is on the ball so that they can execute the movement better so that we can get the reward you know score the points put the ball in the hoop and so it relates to having people practice or maybe get input on your hand was a little bit off here relative to the coach's expectations of where the hand should be but then there's this whole question about visualization and about what dave was saying like playing the music without touching the instrument and the ability that was almost discovered in a bottom-up way of changing beliefs about one's rehearsing mentally but also changing beliefs about performance influencing motor behavior that is more consistent with what we see in active inference and so there's just multiple things that previously would be seen as sort of exceptions or quirks of a pure motor control system like incorporating these deeper elements of our generative model all the way up to even like the psychological like maybe your generative is that your team is fated to lose so then you do choke in that moment because like you know you didn't psychologically want to win or whatever and that's as gina's we're always joking about like the interview with the athletes right after the game it's like i wanted to win or something like that like they're not going to give you that third person my hand you know was on the wrong spot on the ball um it just we're seeing how now as we have just like blue said more computational modeling that recognizes interactionism and that recognizes deep generative models it will be possible to start to slide some of these examples from being like quirky outliers to actually sketching out the structure of inference and action instead of just being seen as sort of like this one weird trick that's going to help you learn free throws it'll be like that's always about your and that includes you as being a team player for example it always is going to include that and then it's going to come down to maybe being modeled better by active inference rather than a reward based framework blue so i just was thinking like you know in your you said you were exploring like the the dynamic you were you were going through the dynamic there was like exploration and then doing something intentionally like learning how to do it and then like perfecting a skill right i was thinking like you left off the end of that like uh just like from a societal like standpoint right like we explore new things like not thinking as an organism but thinking as as a social structure we explore new things we set out the intention to complete those new things we perfect those new things and then we forget them right like so so does that come into play in in this i wonder because i think about scaling active inference a lot like from not in the way that we talked about scaling active inference um in the paper scaling active inference but but just as as from uh like you know small components of a system to systems and you know in that scaling like like are we missing something about about forgetting right like like handwriting if you think about like learning cursive you know that was something like people start out scribbling on paper little kids you know they're playing and they're drawing and whatever and then they they intend to learn cursive they perfect cursive but but like as a society we're not teaching that anymore so they don't know cursive and it's like typing like now we have voice translations so we don't know typing anymore i mean at these types of things like are influential in a society does that come into play here like where i've talked about before you know thinking about doing something like entering the flow state right when we talked about the flow last live stream you know it's like you have the intention to do something and like you think about doing something but when you're really a skilled performer you enter that flow state and you forget about doing it you just do it it's just magically accomplished so is that like flow state comparable to this like forgetting of as a society right so so when you enter that flow state and like you're doing something magically like just with your kinesthetic memory and you're perfect at it does that compare to society like forgetting a task because we've already mastered it like we're we forgot about that we're on to the next thing i wonder if that's relevant anyway sorry if i rambled no it's it's really nice here's what it made me think about was instructions again implicitly or explicitly they have to be remembered or forgotten because if they're not remembered along the chain or at least in the ram of the computer it's like they're never carried out if the instruct if you started a sentence and they forgot what the first part was by the end of the step they can't carry out the instructions so instructions we think about symbolic representation remembering and forgetting but with interactions it doesn't have to be remembered or forgotten it's actually just experienced like in the ant colony the interactions between ants they don't have to be remembered it updates the generative model of each nest mate and then they continue on their path but it's actually just experienced and that's enough in an interactionist framework rather than being remembered i'm not sure if that gets at it i think it's a great point about how as something becomes pervasive and other skills and scaffolds are layered over it there is a forgetting that can happen but also just it's like we don't even need to have memorization of everything like i remember hearing from different college majors and somebody would say somebody would say what major they were and somebody else would say oh well that one always was too much memorization like there was that was too much memorizing and then for that person it wouldn't be it was their major it wouldn't be too much memorizing it just seemed like it was parts of a bigger picture and so they didn't feel like they were memorizing little factoids but rather just interacting with the material not sure dean what do you think about that well blue you didn't say this but you touched on something that's mattered to me for a really long time and that is should we scale bottom up or does it sort of exist in its own realm because that's the thing that you basically said i i i taught for a lot of time and it bothered me greatly that all the learning from from haptic movement was disappearing into a keyboard so so so we topped down that one and then we forgot about it because top down basically erased the bottom up we we seem to think that if we talk down it if we if we give it a nap we can now forget about it and i find that i find that crazy because i've had many quite quite heated debates with people who want to take another bottom-up exercise where we can really learn something and get some joy out of discovery and say well that's i'm going to scale that and i say leave it alone leave it alone it's a bottom-up thing what are you doing i know you can top down it but why can't we just let it explore a little bit and figure things out for itself so dean you really touched on something there with like the haptic movement um and you know i like personally learn super well by writing things down like i take notes i throw the notes away i never look back at them again it's the act of physically writing something down that helps me uh learn um and i mean sometimes i do look back at my notes but but mostly not i just take them to to commit things to memory and um you know my daughter was really struggling with uh learning her times tables right she's like doing these flash cards and the quizlet app and all the stuff and i had her manually copy you know like the times tables and you know instantly like within a month she had them down um and just like destroyed the standardized tests like where they do math computation which is just like rote memorization of your math facts and how fast can you do it she got like 100 and she's you know at off the charts for her grade level in testing so um anyway it's it's really very um real and actual but but in that kinesthetic memory and this goes back to that like motor command you know and like the disappearance of the motor command and so this is like what i think about like these two juxtapositions and i think like there is a space like where you do issue a motor command so like at least for me right like there is a space where it's like okay i need to put my right foot in front of my left foot and you know walk along to learn to balance like i'm issuing a to myself to execute some task but but there are many times like where you do issue that motor command you i mean like think about like if you've ever tried to learn to play the piano like placing your fingers on the keys and stuff and then it just you just do it right like there's no command at that space and so so there's like a mental a kinesthetic like something that happens like where you transition between having to issue the command and not having to issue the command and this is something that i you know i'm excited to talk to the authors about if they are able to join us on a live stream because i because i don't know where is that step where does that step get erased in active inference i mean we aren't always just we aren't always commandless i think i don't know so i'll just put one more one last thing on that if if i think of myself if i think of myself separately from the neck down everything from the neck down is giving me the bottom up my brain is way up here there's things that are supporting that that are giving me that chance to bottom up i don't want those things to disappear i don't want to be a head on a on a cart i don't want to be like that plant with just my head going back and forth between lights right like when do we stop saying that everything has to and like i said you didn't say this but i've had so many people say oh dean if you if you all the stuff that you're doing is great and you you create all these unicorns and all these young people are able to go out and contribute in professional settings but it's not scalable and i go what what do you mean it's not scalable every one of these things is a bottom-up example that it it can happen as long as you don't try to top down it to death interesting the part about the body and neck reminded me of ken wilbur's centaur stage of development which is sort of uh along the integral psychological trajectory there's a step where there it's like half human half horse and so it's like one step beyond kind of a human in dialogue with a horse as two separate entities there starts to be like a fusion in his framework moving towards non-dual framings but there's that stage where it's like uh the top down the bottom up but you know our brain is further out just gravitationally so it makes it seem like it's the top piece but you know what about a person upside down then is it bottom-up signaling well you've already pointed em out he's on the edge of a building back there about ten slides or people on the other side of the world they're upside down right uh and i think uh asking about speech and sign language could be interesting because i think those are gonna be the challenge cases for symbolic the last bastion of instructionism and symbolic messages being passed maybe is happening through language whereas with this sort of elegant skillful performance model maybe you go okay i can see where predictive processing is coming into play where it's not just about value okay so if predictive processing is an improvement from reinforcement learning um or from optimal motor control then i guess i could see why active inference builds upon predictive processing so that's that's the pipeline the active down the rabbit hole pipeline for some but i think when it actually is semantic or symbolic what is being performed that'll be where on one hand there's going to be resistance because it's going to seem the most instructionist but on the other hand i think we're going to tap into the richness of active inference to have a deep generative model including semantic information whereas the semantics are not in this motor control framing so it'll be like double or nothing with for some of the symbolic cases uh blue yeah i just wanted to go back to what um dean was saying earlier about bottom up and um like i don't know uh like using active inference and modeling active inference in like a bottom-up and nested way is super super interesting to me um and and i just i'm i'm wondering like when are we gonna crack that walnut because it's gonna happen i feel like we're we're teetering you know on the edge of that um what will that look like i mean yeah what will it look like or what will be a system where we can explore it in i mean maybe the active in france i think is maybe maybe daniel will grace us with we get to do his paper one day we can do it soon but yeah the only thing i'll add is that we have to people that are participating in this and really are going oh my goodness we're really on the cusp of something just have to be willing to go over the edge of the cliff and i'm not and i'm not joking it's the letting go of all the habits that we formed to get to this place because most people didn't get to we didn't the three of us didn't get here through active inference that came after a whole bunch of things that were more instructionalist and to what to step back it would be much easier than to go over if we've got the prospecting the proposition and the provisions in place we'll go over the edge and it'll all be fine it'll be actually quite thrilling i i really like that like uh sort of that come to the edge there's a different thought chain when you're climbing the ladder to the high jump like we've all jumped off of something that was scary or something like that there's a different thought process when you're climbing maybe because your motor behavior is grounded you're on the ladder or you're on the mountain going up that's one mode that's actually like approach but then right at the end the mind goes blank and you know the mind will say jump jump but there's that resistance and that reminds me one of the few times that i've used vr goggles um it was like you walked into an elevator this was like a scenario walk into an elevator and then it carried you up a building and you could see yourself getting higher and higher up and then the elevator doors opened and you looked down and it was just like dropping off a building and it felt like my foot was just glued to the ground like i felt like i could not take a step forward even though i could inch my foot forward and i could see that it was that i hadn't moved that i was still just on flat ground it was like it just stopped motor behavior to know that it would be resulting in some unpleasant situations in the usual case so dean what do we say to those who are um you know not sure if they want to go to the cliff what do we say to those who are walking towards the cliff and then what do we do with those who are on the edge this is this is fantastic question because this is this is the nut of the of the problem so so do you want do you want instructionalism extended into the active inference realm or do you realize that the whatever identity you take on in terms of with both eyes open with some instructionalism and some interactionism i i don't know what that will be for you or blue or or me i just know it won't be an extension of instructionism alone i don't that's the counter factual here i don't know what that will feel like as different any more than you knew what the difference was going to be in terms releasing to gravity as opposed to pushing against it as you were climbing up those stairs i just know that if you considered it to be just two more stairs as an extension of what you were already doing you're gonna be disappointed so i think it's that that is the that's the big question here as long as you don't as long as when i used to say to kids as long as you don't think that a field is simply an intern outside of school i don't know what that's going to feel like and i certainly don't know what that's going to look like because i'm not out there with you and your sponsor so how can you let go of what you know in order to realize this thing that nobody can give you a direct answer to that's that's a that's a beautiful question daniel because that that for active inference is going to be that's the rule that's going to have to be discovered that that even people who are spending a lot of time focusing on this right now giving it formalisms and and developing the the way to formalize the math are going to have to have themselves otherwise it's just going to be an extension of instructionalism and that would be sad yep it's like jumping off it's not a staircase in reverse right and um i think the idea would be maybe we already have flight suit on where it's like if you climb you climb up with the provisions and then you know when you jump you're gonna be able to do something that was just qualitatively different it's gonna feel different it's going to be different it's going to be a different action and wasn't the point of listening to instructions to change to update to be better or to be and then it's like by being different and following that path we become who we are like somebody said i always wanted to be skilled in this martial art so then they trained along the way to become who they wanted to be and that was them being who they wanted to it's a lot of sort of loose associations but again i think we're working towards making it happen it is it isn't though because instructionalism my point was that the instructions stopped i cannot give you a description of what that will look and feel like so you're going to have to interact with it that's there's not it's not a negotiable if you really want to be there you have to let go of the so i don't think that's a loose association that's a non-negotiable now you may not like it and you may not go there you may step back from the edge and that's fine but don't think that i'm or other people that are actually looking at active inferences as a second eye open won't be aware of what's what you're transfiring whether you're going over or whether you're stepping back that's not a loose association that's that's pretty clear blue any thoughts on that i i guess my question would be let's just stay with this sort of like climbing the instructions to get to the space of interactions what yeah blue god so it just as like climbing the instructions to get to the space of interactions like i feel something is there that is has to do with like the bottom up and the scaling like you know in the systems like we are the instructions as you know parts of a system until we get to to that space as a system the rules well it's like okay we have the ladder going up or you know staircase going up to the jumping off point you could frame that in terms of go to the second step go to the third step go to the fourth step instructionism all the way i mean our interactionism all the way it's like it gives you the rule walk forward so you can replace on a structured ascension you can replace the sequential with a rule or a pattern so that it's interactionism all the way down all the way up and that kind of relates to active as a scale-free framework like bottom up from where blue are you going to start with subatomic or where is the kernel going to be and maybe just interactions all the way down and then at some sufficient level you can granularize or you can coarse grain you can make something look as if it has instructions so i wonder if it's like you know you see this loop right the active inference loop and it's like an oval like this and so i wonder if it's like starts to be like the structure of an atom like okay there's like this oval that's this way and then i wonder if i could superimpose one going the other way a vertical and a horizontal like a little one of these things right you know so that you know because there's this across scales as we scale up and down you know is there a loop going between and then a loop going this way possibly probably i don't know i think about it in that way so this could be the s orbital this is just like a spherical one and maybe there's other trajectories or flows that have different shapes so dave wrote in there and anyone else in this last little bit is welcome to write a comment dave wrote in most effective military units troops are told what to accomplish not what to do so that's the difference for me like push and pull push you know lift the bar is like it's a command about what to do versus like when it's an accomplishment and so it's the teleos or the end directedness that pulls the person and then in that poll there is the space to have the solenoidal flow to to walk straight towards the goal where there's a clear path but then to be like oh yeah but then there's i'm hitting a wall so in order to accomplish i do have to walk around of course that makes sense in the context of being pulled by a goal but that's where the instruction-based robotics always fails because anytime there's something that requires a deviation from you need a total second layer of the model to re-introduce curiosity or epistemic gain into the value framework so then it's this question how do you balance all these second layer bolt-ons so that you can still salvage your value-driven model even when there's aspects that are just clearly better framed within a pragmatic epistemic distinction blue yeah it's like solving the problem without ground truth data right like so how do we you know if we don't know if we've never been exposed to the system before if it's not programmed into our robotic function you know how do we um account for that right so i think that that's where the exploration and the play a lot comes in or maybe tossing and turning like you want to be comfortable and then it just sort of uncoordinated behavior that's just rummaging around searching for something in a box or all these examples where it's so you couldn't say that that twist of your side it was it's not skillful at least for me but it's also not reward driven again you need a way second level bolt-on to be explaining how the rummaging behavior is itself value-driven you can say that there is a value that's pulling or you want the valuable thing in that box but that's we all agree on that we all agree that there's something valuable in the box and that search behavior is going to be required are we going to have a process theory that allows for exploration and then zeroing in on a goal or will we have a process theory that tries to go like well if the end is valuable then let's just back propagate value yeah or we don't know if we don't know what's valuable in the box right something is valuable in here like how do we find out what is it that's valuable okay so we implement the search but then exploring what is the valuable item out of you know 50 items in the box right yep it's almost like in the simplest cases uh the value learning is defensible when we have this hill-climbing robot on a single hill okay but now you got a triple bottom line you got the financial success but you also have environmental and social considerations of a business so will you reduce all three of them to one of them or try to generate a fourth standard that you're going to reduce them all like a common currency for all of your values and rewards or could there be a deep generative model where the organization has preferences in the financial social and environmental realm and then take stock of its action possibilities not even described in the value mindset maybe every time you have a meeting it's like square zero what can we do whereas in active inference we start with the space of policies the affordances that the niche provides and so you don't need to consider policies that cannot be done and you should re-weight policies accordingly to how likely or tractable they are so there's just so many ways that we see that the mathematical framings get expanded on through uh relaxation of some of their attributes just like first in 2011 wrote which is that optimal control can be cached as active inference with three simplifications and then he walked through those which we discussed a little bit last time but this is kind of cool how we're building out some differences between the motor control approaches that have been used in the past and active inference as a process theory and then the way that that encompasses what we've seen before and then maybe also opens up the into just areas that weren't even considered like we can add in affordances plus preferences that i think that embodied skillful and an active inference implies a huge leap of faith back to that uh jumping off the because it is i mean that's a that's a cliche that people can probably appreciate what do you think the trust or faith is yeah that's it yourself yourself right or through it act or act not there is no infer no there is there is um so it's interesting yeah how how can we distill the similarities and differences so that when we see one of those limitations being like okay we had um uh curiosity slash pragmatic plus epistemic so that when we see a sort of here's the new hack on value we can see that as a saddle point for where that model could move towards an active framing just by kind of rewiring some of the modules that it has um here's a great question from the live chat bill wrote would a priority parameter according to depletion levels of different resources applicable yeah that's back to the pools thing yeah i would say yes okay dean with the pools here would be my uh thought from the multi-scale decision making let's just think about a bee colony the resources that they need to forage for include like pollen which has a lot of protein and fat and it's helpful for the developing larva then nectar which is like high octane gasoline a lot of carbs good for the adults um and then depending on the environment they might need like water or salt or other different resources so the depletion of uh there might be an imbalance of the nutrients and you can think about that similarly for an organism like rabbit starvation when you only eat protein your body will enter starvation mode because you know other fats and stuff are needed so definitely multiple resources are being balanced and part of the challenge of decision making under just as we've been getting at is that there's not just one kind of outcome like it's enough to be uncertain about value but now when we're thinking about these multi-dimensional resource landscapes whether it's just protein versus carbs or whether it's a lot more nuanced like the triple bottom line we really do need a principled way of talking about the relative depletion different uh resources so in the bee colony case selection has shaped nest mates so that they use the rate and the type of interactions they have in order to update their nest mate level behavior so i'm having a lot of interactions with hungry larva maybe we need more pollen i'm having a lot of interactions with food reservoirs maybe that can slow me down because we probably have enough forage for now so colonies that don't consist of nest mates with that productive way of interpreting their interactions not instructions those colonies are swept off the table and so we end up seeing distributed systems succeed where they act as if there is a correct prioritization amidst uncertainty of different kinds of and then if we use active inference instrumentally in the deep generative model of the body's nutrient ratios or the colonies relative balance between protein and carbs then it's as if there is a priority parameter where when something is depleted it's the one that's prioritized but we don't need that expressively in we can have a multi-dimensional um generative model of different resources and then actions will be selected according to their expected free energy being minimized almost going along the curve that takes us on that edge of explore and exploit from wherever we are in that resource depletion landscape towards our preferences so would be totally open to hearing other explanations or maybe there is a priority parameter that can come into but one sort of longitud response would be organisms do act adaptively to prioritize depleted resources but that doesn't mean that there needs to be a parameter in our model that explicitly prioritizes specific resources it's enough just to have a preference vector that includes multiple types of resources blue no i think that that's right i think the right so you can make it as simple or complicated as you want but i always think less parameters in the model is better it just makes me think you know how do we get those preferences and then what happens when the niche changes so that preferences that were adaptive like to act as if prioritization of sugar and salt over all other preferences for example what happens when the situation changes so how do we integrate high precision priors with highly accurate sensory input that's something we've talked about before how do we get the best of both worlds with being able to like fly blind because we have a deep generative model but then also when there is sensory input that requires a categorical shift in cognitive model for example we want have the fluidity to do that and we don't want to be wasting our time in one mode when it's the wrong mode we don't want be hard when hardness is a flaw or be soft when softness is a flaw so i wonder if there's like um you know i mean for organisms like bees and birds like there's some programmed i mean things are shifting all the time right like there's you know a huge amount of like migratory stuff that comes into play and hibernating and all these things right um and so like for us like we noticed that prioritized like shift like i need water right now like you know if we're suddenly like parched right like the need for water is overwhelming but there's not this like because we have offloaded so much of our needs into our niche like the need for to stay warm or whatever we build houses we wear clothes so we've offloaded a lot of this stuff so i i wonder if it's like um the instinct becomes less instinctual i wonder if that's if that happens like the instinct for to sequester food or fly south for the winter or these types of things um just because you know we don't um we don't we've just offloaded our need to do those things so it just becomes i don't know there i think that's another uh distinction here here we can include extended slash embedded you know four e's seventies eleven e's however many there are um we can at least frame those situations in terms of niche modification or niche construction where is that in motor control it's nowhere where is golf club with a squishy handle that is being you know interacted with as an we don't have those nice interfaces whereas active inference has excellent interfaces quantitatively and qualitatively for being enriched by pieces that are as dean would say maybe non-negotiable for the actual creature but it'd be totally ad-hoc where would you put the instrumentation in here is it a state estimator should i be estimating where my golf club is or is it enough just to estimate my hands and the angle of the hands and then do another second level prediction not quite clear you could probably implement it multiple ways and here's another nice comment from dave so dave said the term in cybernetics so it's always good to connect back to related areas or putting drives into a model is algodonic so here's the definition of algodonic comes from i guess algos meaning pain and hedon meaning pleasure sorry dave for the mispronunciation um putting multiple algodonic drives like pains and pleasures may yield complex instructive behaviors but there are so many advantages in having a model i think active inference so sophisticated that it generates its drives the drives come out of the model rather than being put in so yes and he wrote then clarifying i mean the innumerable drives emerge from the model's internal dynamics if your robot is spontaneously curious spontaneously attends to events or responsive externals it's as if it has a curiosity parameter so it's true maybe we could even write like drives quote you know as if i'm gonna i'm gonna take shakespeare thanks hamlet you are great a great prince um drives drives emerge from model um don't need to be put in so instead of uh like maybe the rapid being put in the hat this is like the rabbits emerge from the hat if you you know put in enough hydrogen and enough time so that's interesting to think about where cybernetic models come into play here so let's give just a few final minutes all right any final questions in the chat and thanks of course to dave and to phil for the great questions we can just think about um maybe dot one dean is um being on the edge dot zero is climbing and so it's a two directional trail you know there's people going up to half dome and there's people coming down so dot zero is two directional highway lots of on-ramps and off-ramps but we try to make the topics accessible for those who are not in the active inference game but they're related they're familiar with related topics like the keywords usually and in the dot zero we're going to go from bridging the keywords and the topics of broader concern to active inference so that's the on-ramp to active inference but then also for those who already in the active inference mode of learning by doing and working then it's their window out to like oh wow you know this idea that i had been working active inference in one context it's also applying outwards so that's in the dot zero dot one you who whoever and whenever it is uh we are in the the bathtub face the part between the book ends that's we're at the edge and we can look back to the dot zero and see again recapping the the steps that we took but also we want to start looking out and then i don't know where dot two would be but dot two is when we sort of take the next step and the dot two is the no going back i mean i i can add a twist to the fact that i'm i'm going to be hitting the water soon and that's a skillful embodied movement right we actually watch people that do the 10 meter platform and can add up quite a few things before they hit the water and regain control but i i i i think this is this is what interactionism is versus i mean the way that this is set up the zero one two is interactionism it's more about this than it is about um bi-directionality and i think you talked about that at the beginning of today's episode right we have a a structure that we hold to a rail that we could modify as well but we hold to the rail so that we don't have to frame it like um we weren't instructed to do this and i don't know i hope that people don't feel like these are instructions but rather interactions or engagements with them and then just like so many other things that we're seeing online the new technological affordances allow it to actually be an interaction we can have people asking questions live being a part of the conversation whereas if it were just recording a video and releasing it or record making a book and releasing it it'd be hard to break out of the instructionist frame because the deliverable would be static and very unidirectional here we have a different affordance that helps us break out of that frame basically for free because even if instructions are being conveyed on the live stream it would still be within an interaction of participants of individual and collective and of the community so any final thoughts on one no fun times thanks both of you for the the dot zero and dot one fun we've had a few hours to think about it but you know the paper is the distillation of many more hours so there's always value dare i say and going back and playing with it because it's fun and it's just yep there are great topics to be considering about and learning so cool thanks everyone for watching thanks dean and blue and we'll see you next week for 23.2 thanks guys bye